Characterization of the Teledyne FLIR Boson 320 Uncooled Microbolometer Camera

Lab Report and Experiment Protocol

NETD Measurement and Absolute Temperature Accuracy Measurement

Device Under Test: FLIR Boson 20320A034-6IARX (Industrial grade, radiometric, 34° lens)

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Adapted from: Krempel, L. (2020), "Thermal Infrared Imager for Space Applications", TU Munich master's thesis — Reproducible Experiment Protocol (Fraunhofer IMS prototype). This document has been restructured and rewritten for the Teledyne FLIR Boson core, which differs substantially in architecture, calibration interface, and available sensors.

Contents

1. Introduction and Scope

This lab report describes a reproducible test procedure to characterize two key performance parameters of an uncooled long-wave infrared (LWIR) camera core: the **Noise Equivalent Temperature Difference (NETD)** and the **absolute temperature measurement accuracy**. The procedure is adapted from a Fraunhofer/TU Munich master's thesis protocol originally developed for a custom prototype LWIR imager, and has been rewritten to match the capabilities and constraints of a commercial-off-the-shelf Teledyne FLIR Boson camera core.

The instructions are written so that a reader with no prior background in infrared imaging or radiometry can perform the experiments. Section 2 introduces all technical terms and abbreviations used later in the document. No detector-temperature control is required or used in this version of the protocol, since the Boson is an uncooled device with a fixed internal operating temperature range.

This document covers exactly two experiments:

- **Experiment 1 — NETD Measurement:** determines the smallest temperature difference the camera can resolve above its own noise floor.
- **Experiment 2 — Absolute Temperature Accuracy Measurement:** determines how closely the camera's reported scene temperature matches the true, independently known temperature of a target.

2. Background Concepts and Glossary

This section briefly explains every specialized term, abbreviation, and concept used in the rest of the report. Readers already familiar with infrared imaging may skip to Section 3.

2.1 Microbolometer and Uncooled Detectors

A **microbolometer** is a tiny thermal sensor whose electrical resistance changes when its temperature changes. In an infrared camera, each pixel is a microbolometer that heats up slightly when infrared (heat) radiation from the scene falls on it. An **uncooled** detector operates near ambient temperature, without cryogenic cooling, unlike older or more expensive "cooled" infrared detectors. The Boson is an uncooled microbolometer array, meaning its own operating temperature drifts with ambient conditions and self-heating, which is one reason careful calibration (FFC, see 2.4) is required.

2.2 Radiometry and Temperature-Linear Output

Radiometry is the measurement of electromagnetic radiation, in this context, converting the raw electrical signal of each pixel into a physical unit such as scene temperature. A **radiometric** or **radiometry-capable** camera can report an actual temperature value for each pixel, rather than just a relative brightness. The Boson supports two radiometric output modes: **temperature-stable (T-stable)** output, which is proportional to incoming radiation only, and **temperature-linear (T-linear)** output, in which the 16-bit pixel value is directly proportional to the absolute scene temperature in Kelvin. This report uses T-linear output so pixel values can be read out directly as temperatures.

2.3 Blackbody and Emissivity

A **blackbody** (or black-body emitter) is a calibrated reference source that radiates infrared energy very predictably according to its set temperature. **Emissivity** is a number between 0 and 1 describing how efficiently a surface radiates heat compared to a perfect blackbody (emissivity = 1). Real-world blackbody sources used in labs typically have an emissivity around 0.95–0.99, which is high enough to be treated as a near-ideal reference. Emissivity must be set correctly in the camera's radiometry settings, or absolute temperature readings will be biased.

2.4 Flat-Field Correction (FFC)

Flat-Field Correction (FFC) is a self-calibration process in which the camera briefly views a uniform thermal scene (using its internal shutter) and recalculates per-pixel correction values so that a uniform scene produces a uniform image. FFC compensates for pixel-to-pixel variation and for drift in the camera's own internal temperature. It takes less than one second and can be triggered automatically, by a timer, or manually via software command. In this report, FFC is always triggered manually via the Boson SDK immediately before each measurement, to guarantee a known, fresh calibration state.

2.5 NUC (Non-Uniformity Correction)

Non-Uniformity Correction (NUC) is the broader set of factory-calibrated correction terms (of which FFC is the runtime-updated part) that compensate for pixel-to-pixel manufacturing variation, lens shading, and temperature effects. NUC tables are pre-calibrated at the factory for different camera temperature ranges and gain states, and are loaded automatically by the camera.

2.6 AGC (Automatic Gain Correction) and Raw Data Taps

The Boson's internal signal pipeline produces data at multiple points, or "taps". The **pre-AGC tap** provides raw 16-bit data proportional to (in this case) scene temperature. The **post-AGC tap** provides an 8-bit, contrast-enhanced image intended only for human viewing on a display, and is not suitable for quantitative measurement. **AGC (Automatic Gain Correction)** is the algorithm that performs this 16-bit to 8-bit contrast enhancement. All measurements in this report use the 16-bit pre-AGC data tap, obtained through the Boson Software Development Kit (SDK).

2.7 NETD (Noise Equivalent Temperature Difference)

NETD (also written NEDT) is the smallest scene-temperature difference that produces a camera output change equal to the camera's own random (temporal) noise. In simple terms: it is the smallest temperature difference the camera can reliably "see" above its own electronic noise. It is usually expressed in milli-Kelvin (mK). A lower NETD means better thermal sensitivity. The Boson 320 Industrial grade used in this test is specified by the manufacturer at less than 40 mK.

2.8 Absolute Temperature Accuracy

While NETD describes how small a temperature *difference* the camera can detect, **absolute temperature accuracy** describes how close the camera's reported temperature value is to the true, independently known temperature of the target. Even a camera with excellent NETD can have a systematic offset or error in its absolute readings, which is what this second experiment quantifies. The Boson's datasheet specifies this accuracy as approximately ±3°C to ±5°C depending on operating conditions.

2.9 Gain States (High-Gain / Low-Gain)

The Boson can operate in **high-gain** state (best sensitivity/lowest NETD, but limited maximum scene temperature before saturation) or **low-gain** state (lower sensitivity but a much wider temperature range). For this protocol, the camera should remain in a single, fixed gain state (high-gain is recommended for the NETD test) so that results are not confused by automatic switching mid-measurement.

2.10 Abbreviations Reference

Abbreviation	Meaning
NETD / NEDT	Noise Equivalent Temperature Difference
FFC	Flat-Field Correction
NUC	Non-Uniformity Correction
AGC	Automatic Gain Correction (contrast enhancement for display)
SDK	Software Development Kit
FPA	Focal Plane Array (the detector chip itself)

ROI	Region of Interest
T-linear	Temperature-linear radiometric output mode
T-stable	Temperature-stable (flux-linear) radiometric output mode
FOV	Field of View
LWIR	Long-Wave Infrared (8–14 micrometer band)

3. Device Under Test and Available Equipment

3.1 Device Under Test

Property	Value
Model	Teledyne FLIR Boson 20320A034-6IARX
Sensor format	320 x 256 (QVGA)
Pixel pitch	12 micrometers
Spectral range	8-14 micrometers (LWIR)
Lens	34.1° x 27.3° HFOV, f/1.01, 6.3 mm focal length
Camera grade	Industrial (< 40 mK typical NETD, > 99% operable pixels)
Radiometry	Capable (R variant); supports T-stable and T-linear output
Shutter	Integrated (internal, used for automatic/manual FFC)
Cooling	Uncooled microbolometer, no user detector-temperature control

3.2 Manufacturer-Rated Specifications (for later comparison)

Parameter	Rated value
NETD (Industrial grade, Boson)	< 40 mK
Absolute radiometric accuracy (high-gain, 25°C ambient)	+/- 3 C (for 10-50 C scene), +/- 3 C (for 100 C scene)
Absolute radiometric accuracy (high-gain, 50°C ambient)	+/- 5 C (for 10-50 C scene), +/- 4 C (for 100 C scene)
Test conditions for rated accuracy	Steady-state (FPA drift < 0.1 C/min), target in central ROI, FPA within 0.2 C

Source: Teledyne FLIR Boson Product Datasheet, Rev 350 (Tables 1, 20).

3.3 Available Lab Equipment

- Teledyne FLIR Boson 20320A034-6IARX camera core with USB interface to a host PC
- Boson SDK / Boson App (software) for camera configuration, FFC triggering, and 16-bit frame capture
- Calibrated temperature source with a flat radiating target, placeable directly in front of the lens
- Black-body emitter (independent reference source, adjustable and stable set-point)
- General lab equipment: optical bench or stable mount, stands, cabling, host PC

No external thermometer, no detector-temperature control (Peltier) system, and no external shutter or shutter-mounted temperature sensor are available. All ground-truth temperature values in this protocol therefore come from the calibrated temperature source and the black-body emitter only.

4. Differences From the Original Protocol

The original protocol (Krempel, 2020) was written for a custom laboratory prototype with a separate detector, external motorized shutter with its own digital thermometer, Peltier-based detector temperature control, and direct access to raw ADC counts via a custom readout board. The Boson is a sealed, commercial imaging core with a different set of capabilities. The table below summarizes how each original concept maps onto the Boson setup.

Original protocol	Boson-adapted equivalent
Detector temperature set via Peltier stage	Not available; camera runs at its natural, uncooled operating temperature. A
Manual Vref1 half-scale tuning before each run	Manual Flat-Field Correction (FFC) triggered via SDK before each run
External shutter with DS18B20 temperature sensor as low-temperature reference	No external shutter; the calibrated temperature source (placed as a t
Raw 16-bit ADC counts via custom readout board	16-bit pre-AGC data tap accessed via Boson SDK
Black-body-only or shutter-only reference scenes	Calibrated temperature-source target and black-body emitter used together t
30-second validity window for offset calibration	FFC validity governed by Boson's own FFC-desired / temperature-delta flag

5. Common Setup Procedure

Both experiments share the same physical setup and startup sequence. Complete all steps in this section before beginning either experiment, and repeat the FFC step (5.4) immediately before every data-capture run.

5.1 Physical Setup

1. Mount the Boson camera on a stable stand or optical bench.
2. Place the calibrated temperature source's target directly in front of the lens, centered on the optical axis, at a distance beyond the lens's minimum focus distance (see camera/lens documentation; for the 6.3 mm/34° lens, avoid placing the target closer than approx. 20 cm, to prevent blur).
3. Position the black-body emitter so it can be swapped into the same position/orientation as the calibrated source with minimal disturbance to the camera or alignment (e.g., on a small slide rail or turntable).
4. Ensure the target (source or black body) fills the full field of view, or at minimum, fully covers the central Region of Interest (ROI), since radiometric accuracy is only specified for pixels within the central ROI.
5. Connect the Boson to the host PC via USB and confirm it is detected by the Boson SDK/App.

5.2 Camera Configuration

1. Power on the camera and allow the standard boot sequence to complete (approx. 2.5 seconds).
2. In the SDK/App, enable Radiometry and select Temperature-Linear (T-linear) output mode, so that pixel values correspond directly to scene temperature.
3. Set the environmental factors (see Section 2.3 and 2.4 of the Boson datasheet) as accurately as possible: target emissivity (use the black body's or calibrated source's rated emissivity value), atmospheric transmission (approx. 1.0 for short lab distances), atmospheric temperature (room temperature), and background temperature (room temperature).
4. Set the Gain Mode to a fixed state — High-Gain is recommended for both experiments, since the scene temperatures used are expected to remain below approx. 100 C (see high-gain radiometric accuracy range, Table 20).
5. Set FFC Mode to Manual, so that no automatic FFC events occur during a data-capture run.
6. Configure the video/data tap to output pre-AGC 16-bit data (not the 8-bit AGC/display stream).

5.3 Thermal Warm-Up

Because the Boson is uncooled and has no user temperature control, allow the camera to run continuously for at least 15-20 minutes after power-up before taking any measurement. This lets the camera pass through its initial rapid self-heating phase (the first 90 seconds after power-up trigger extra-frequent automatic FFC events even in automatic mode) and reach a steady thermal state where its own temperature drifts slowly (below approximately 0.1 C per minute).

Record the room's ambient temperature at the start of each session. This is your only substitute for detector-temperature control, and should be reported alongside all results.

5.4 Manual Flat-Field Correction (FFC) Procedure

This step replaces the original protocol's Vref1 reference-voltage procedure, and must be performed immediately before every data-capture run described in Sections 6 and 7.

1. Ensure the camera has completed its thermal warm-up (Section 5.3).
2. Send the manual FFC command via the SDK. The internal shutter will close briefly (less than one second) and the camera will update its internal correction terms.
3. Confirm via the SDK/telemetry status flags that the FFC State reports 'complete' before proceeding.
4. Do not move the camera, target, or change any camera settings between the FFC command and the start of data capture.

6. Experiment 1: NETD Measurement

6.1 Purpose

This experiment measures the camera's Noise Equivalent Temperature Difference (NETD): the smallest scene temperature difference the Boson can resolve above its own random (temporal) pixel noise. The result will be compared against the manufacturer's rated value of < 40 mK for the Industrial grade Boson.

6.2 Principle

Two scenes of known, slightly different temperatures are imaged in sequence, without recalibrating the camera in between. For each scene, many consecutive frames are captured so that the per-pixel temporal (frame-to-frame) noise can be measured as the standard deviation across frames. The known temperature difference between the two scenes, divided by the ratio of signal difference to noise, gives the NETD.

6.3 Materials Needed

- Full common setup (Section 5), camera warmed up and configured for T-linear, high-gain, manual FFC
- Calibrated temperature source, set to a known first temperature T1
- Calibrated temperature source (or black body) set to a known second temperature T2, ideally 2-10 C different from T1

> Using two close-spaced setpoints on the same calibrated temperature source (rather than the black body for both, or the black body vs. the internal shutter as in the original protocol) avoids the black body's approx. 30-minute thermal settling time being incurred twice within a single measurement, and avoids relying on an internal shutter temperature sensor that the Boson does not expose to the user.

6.4 Procedure

1. Complete the common setup (Section 5), with the calibrated temperature source's target filling the central ROI.
2. Set the calibrated temperature source to T1 and allow it to stabilize (follow the source's own settling-time specification).
3. Perform a manual FFC (Section 5.4).
4. Capture 50 consecutive 16-bit frames while the camera views the T1 scene. Do not move the camera or target, and do not trigger another FFC.
5. Without changing any camera settings and without triggering FFC, change the calibrated source's setpoint to T2 and allow it to stabilize.
6. Capture another 50 consecutive 16-bit frames while the camera views the T2 scene.
7. Record the exact values of T1 and T2 as reported by the calibrated source, and the ambient room temperature.

6.5 Data Analysis

For each of the two 50-frame data sets, and for each pixel within the central ROI:

1. Compute the mean pixel value across the 50 frames (this gives the signal level for that scene).
2. Compute the standard deviation of the pixel value across the 50 frames (this is the temporal noise for that scene).

Then compute, per pixel (or averaged over all central-ROI pixels):

NETD = (T2 minus T1) multiplied by (average temporal noise) divided by (mean signal at T2 minus mean signal at T1)

Since T-linear output already reports temperature directly (Kelvin x 100 in high-gain state), the 'signal' terms above can be taken directly in temperature units, simplifying the calculation compared to the original ADC-count-based formula.

6.6 Things to Watch Out For

- Keep the scene completely static during both 50-frame captures; any motion reintroduces apparent noise and will inflate the measured NETD.
- Do not let T1 and T2 differ so much that the camera's automatic gain-state logic would want to switch gain states; keep both well within the high-gain range (below approx. 100 C, per Table 20 test conditions).
- Confirm the FPA (camera) temperature has not drifted by more than 0.2 C since the last FFC (visible via telemetry/status), matching the datasheet's own accuracy test condition.
- Avoid touching or breathing near the calibrated source or camera lens during capture, as this introduces stray thermal signals.
- Repeat the full measurement at least 3 times to check repeatability, and report the average and spread of the NETD result.

7. Experiment 2: Absolute Temperature Accuracy Measurement

7.1 Purpose

This experiment quantifies how closely the Boson's reported scene temperature (via T-linear radiometric output) matches the true, independently known temperature of a target, across a range of temperatures. The result is compared against the manufacturer's rated accuracy of approximately +/- 3 C to +/- 5 C, depending on ambient and scene temperature (Tables 20-22 of the Boson datasheet).

7.2 Principle

A target of precisely known temperature is placed in the camera's field of view, and the Boson's own reported temperature reading for that target is recorded. This is repeated across a range of temperatures using both the calibrated temperature source and the black-body emitter as two independent references, and the difference between the camera's reading and the true reference temperature is plotted and analyzed.

7.3 Materials Needed

- Full common setup (Section 5), camera warmed up and configured for T-linear, high-gain, manual FFC
- Calibrated temperature source, steppable through a range of known temperatures
- Black-body emitter, steppable through a range of known temperatures, as an independent cross-check reference

7.4 Procedure

1. Complete the common setup (Section 5).
2. Perform a manual FFC (Section 5.4).
3. Set the calibrated temperature source to the first target temperature (suggested test points: 10 C, 25 C, 50 C, and 70 C, spanning the datasheet's tested range of 10-100 C in high-gain state).
4. Allow the source to fully stabilize per its own settling-time specification, and confirm the FPA temperature has not drifted by more than 0.2 C since the last FFC.
5. Record the Boson's reported spot-meter or ROI-mean temperature for the target (using the central ROI, per Section 11.5.1 test conditions of the datasheet).
6. Repeat for each planned test temperature on the calibrated source, re-triggering FFC only if the FFC-desired flag is raised or FPA drift exceeds 0.2 C.
7. Swap in the black-body emitter and repeat the same sequence of temperature setpoints, allowing its longer settling time (up to approx. 30 minutes per setpoint change) as an independent cross-check of the calibrated source's results.
8. Record ambient room temperature throughout, since accuracy is ambient-temperature dependent (Table 20).

7.5 Data Analysis

1. For each test point, compute the deviation: Boson-reported temperature minus true reference temperature.
2. Plot Boson-reported temperature (y-axis) against true reference temperature (x-axis) for both the calibrated source and black-body data sets on the same graph.
3. Plot the deviation (error) as a function of true temperature, and compare it against the datasheet's rated accuracy band (e.g., +/- 3 C at 25 C ambient, +/- 5 C at 50 C ambient, see Table 20).
4. Compare the calibrated-source-derived curve and the black-body-derived curve; large discrepancies between the two indicate either an emissivity-setting error or a positioning/alignment issue.

7.6 Things to Watch Out For

- Always verify the emissivity setting in the camera's radiometry configuration matches the actual emissivity of whichever reference (calibrated source vs. black body) is currently in view; a mismatch here is the most common source of large systematic errors.
- Keep the target centered in the field of view, filling the central Region of Interest, since the datasheet's accuracy specification applies only there — accuracy is worse toward the edges of the frame.
- The black body requires up to approximately 30 minutes to settle after each setpoint change; do not record data before it has fully stabilized.
- Avoid placing the target closer than the lens's minimum working distance, or the image will blur and the ROI may include out-of-focus background.
- Re-run the manual FFC whenever the FFC-desired flag appears or the FPA temperature has drifted by more than 0.2 C, and note in your data log whenever this happens.
- Do not compare results taken at very different ambient temperatures without accounting for the ambient-dependence shown in Table 20 of the datasheet.

8. Data Recording Templates

8.1 Experiment 1 (NETD) Log Sheet

Field	Value
Date / time	
Ambient temperature (C)	
Warm-up time (min)	
Gain mode	High-Gain
T1 setpoint (C)	
T2 setpoint (C)	
FFC performed before run? (Y/N)	
Mean signal at T1 (central ROI)	
Std. dev. (noise) at T1	
Mean signal at T2 (central ROI)	
Std. dev. (noise) at T2	
Computed NETD (mK)	
Rated NETD (mK)	< 40 mK
Notes / anomalies	

8.2 Experiment 2 (Absolute Accuracy) Log Sheet

True Temp. (C)	Source	Boson Reading (C)	Deviation (C)	FFC redone? (Y/N)	Notes
10	Calibrated source				
25	Calibrated source				
50	Calibrated source				
70	Calibrated source				
10	Black body				
25	Black body				
50	Black body				
70	Black body				

Ambient temperature during this session (C): _______________

9. Safety and General Precautions

- Black-body emitters can reach surface temperatures that are hot to the touch; avoid direct skin contact and allow adequate cool-down time before handling.
- Do not point the camera lens directly at very high-intensity heat sources beyond the specified scene-temperature range, as this can saturate or, in extreme cases, damage the sensor.
- Handle the Boson core carefully; it is a bare OEM module without a protective housing and is sensitive to ESD (electrostatic discharge). Use grounding precautions when handling.
- Allow the black body's full settling time (up to 30 minutes) after each setpoint change before recording data, both for accuracy and to avoid thermal shock to the unit.
- Keep cabling and connections secure and strain-relieved on the optical bench to avoid accidental disconnection during multi-frame captures.

10. References

1. Krempel, L. (2020). *Thermal Infrared Imager for Space Applications*. Master's thesis, Technical University of Munich. Reproducible Experiment Protocol (Fraunhofer IMS prototype).

2. Teledyne FLIR. *Boson and Boson+ Thermal Imaging Core Product Data Sheet*, Doc. # 102-2013-40, Release 350, Rev 3.0.26605, Publication Date June 20, 2022.